

Elementary Problems Polynomial Solutions

Version 1.0

2-9-2018

Basic Work

1. **Solution 1:**

The equation can be factorized as

$$(x + 3)(2x + 1) = 0.$$

This yields $x + 3 = 0$ or $2x + 1 = 0$. Thus, the solutions are $x = -3, -\frac{1}{2}$.

Solution 2:

Using the quadratic formula, we have

$$x = \frac{-7 \pm \sqrt{7^2 - 4(2)(3)}}{2(2)} = -3, -\frac{1}{2}.$$

2. The discriminant is $\Delta = (-2)^2 - 4(3)(6) = -68$. As this is negative, there is no real solution.

3. If $a = 0$, then the equation is linear, and hence it has only one solution.

If $a \neq 0$, then the discriminant is

$$\Delta = (-4)^2 - 4(a)(a) = 16 - 4a^2.$$

The equation has two distinct real solutions if and only if $\Delta > 0$, i.e. $16 > 4a^2$. This holds when $-2 < a < 2$.

Therefore, a is any nonzero real number such that $-2 < a < 2$.

4. Note that $x = -1$ is a solution. Thus, the equation can be factorized as

$$(x + 1)(x^2 + 4x + 4) = 0.$$

This can be further factorized as

$$(x + 1)(x + 2)^2 = 0.$$

The solutions are $x = -1, -2$.

5. **Solution 1:**

Let $x = \sqrt{3}y$. The equation becomes

$$6\sqrt{3}y^3 - 3\sqrt{3}y^2 - \sqrt{3}y - 2\sqrt{3} = 0,$$

i.e.

$$6y^3 - 3y^2 - y - 2 = 0.$$

Note that $y = 1$ is a solution. Thus, this can be factorized as

$$(y - 1)(6y^2 + 3y + 2) = 0.$$

This yields $y = 1$ or $6y^2 + 3y + 2 = 0$. The latter equation has no real solution since $\Delta = 3^2 - 4(6)(2) = -39 < 0$. Hence, the only solution is $y = 1$, i.e. $x = \sqrt{3}$.

Solution 2:

Note that $x = \sqrt{3}$ is a solution. Thus, $x - \sqrt{3}$ is a factor. The equation can be factorized as

$$(x - \sqrt{3})(2x^2 + \sqrt{3}x + 2) = 0.$$

This yields $x = \sqrt{3}$ or $2x^2 + \sqrt{3}x + 2 = 0$. The latter equation has no real solution since $\Delta = (\sqrt{3})^2 - 4(2)(2) = -13 < 0$. Hence, the only solution is $x = \sqrt{3}$.

6. Note that $x = 1$ is a root. The equation can be factorized as

$$(x - 1)(x^2 + (a + 2)x + (3a - 2)) = 0.$$

The equation has 3 distinct real roots if and only if $x^2 + (a + 2)x + (3a - 2) = 0$ has 2 distinct real roots other than 1. Firstly, this quadratic equation has 2 distinct real roots if and only if the discriminant is positive, i.e.

$$\Delta = (a + 2)^2 - 4(1)(3a - 2) > 0.$$

This is equivalent to $a^2 - 8a + 12 > 0$, which can be factorized as

$$(a - 2)(a - 6) > 0.$$

This shows $\Delta > 0$ if and only if $a > 6$ or $a < 2$. Next, 1 is a root of

$$x^2 + (a + 2)x + (3a - 2) = 0$$

if and only if $1 + (a + 2) + (3a - 2) = 0$, i.e. $a = -\frac{1}{4}$. It follows that the cubic equation has 3 distinct real roots if and only if $a > 6$ or $a < 2$ together with $a \neq -\frac{1}{4}$.

7. By the rational root theorem, for any rational root $\frac{m}{n}$ of the equation in simplest form, m must divide the constant term 24 and n must divide the leading coefficient 1. Thus, there are only a few possible rational roots. After some trials, one may note that $x = 2, -2, -3$ are solutions. The equation can hence be factorized as

$$(x - 2)^2(x + 2)(x + 3) = 0.$$

This shows $x = 2, -2, -3$ are the only solutions.

8. By the rational root theorem, the only possible rational roots are $\pm 1, \pm 2$. One may check that none of these is a solution. Thus, there is no linear factor with rational coefficients.

We consider the factorization $x^4 + x^3 + x^2 + 2 = (x^2 + ax + b)(x^2 + cx + d)$. Expanding and comparing coefficients, we need

$$\begin{aligned} a + c &= 1, \\ ac + b + d &= 1, \\ ad + bc &= 0, \\ bd &= 2. \end{aligned}$$

If we look for integer solutions, then we may start with $(b, d) = (\pm 1, \pm 2)$ (note that the order does not matter). Afterwards, it is not hard to solve the equation and get one possible factorization

$$x^4 + x^3 + x^2 + 2 = (x^2 - x + 1)(x^2 + 2x + 2).$$

Therefore, we have $x^2 - x + 1 = 0$ or $x^2 + 2x + 2 = 0$. The discriminants of both quadratic equations are negative. Hence, there is no real solution.

9. Clearly, $x \neq 0$. Dividing both sides by x^2 , we obtain

$$x^2 - 2x - 22 - \frac{2}{x} + \frac{1}{x^2} = 0.$$

Let $y = x + \frac{1}{x}$. Then $y^2 = x^2 + \frac{1}{x^2} + 2$. Therefore, the above equation becomes

$$y^2 - 2y - 24 = 0.$$

This can be factorized as $(y - 6)(y + 4) = 0$. The solutions are $y = 6, -4$.

For $x + \frac{1}{x} = 6$, we get $x^2 - 6x + 1 = 0$. Using the quadratic formula, the solutions are $x = 3 \pm 2\sqrt{2}$.

For $x + \frac{1}{x} = -4$, we get $x^2 + 4x + 1 = 0$. Using the quadratic formula, the solutions are $x = -2 \pm \sqrt{3}$.

So the solutions are $x = 3 \pm 2\sqrt{2}, -2 \pm \sqrt{3}$.

10. Clearly, $x \neq 0$. Dividing both sides by x^2 , we obtain

$$x^2 - x - 4 + \frac{1}{x} + \frac{1}{x^2} = 0.$$

Let $y = x - \frac{1}{x}$. Then $y^2 = x^2 + \frac{1}{x^2} - 2$. Therefore, the above equation becomes

$$y^2 - y - 2 = 0.$$

This can be factorized as $(y - 2)(y + 1) = 0$. The solutions are $y = 2, -1$.

For $x - \frac{1}{x} = 2$, we get $x^2 - 2x - 1 = 0$. Using the quadratic formula, the solutions are $x = 1 \pm \sqrt{2}$.

For $x - \frac{1}{x} = -1$, we get $x^2 + x - 1 = 0$. Using the quadratic formula, the solutions are $x = \frac{-1 \pm \sqrt{5}}{2}$.

So the solutions are $x = 1 \pm \sqrt{2}, \frac{-1 \pm \sqrt{5}}{2}$.

11. Let

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

be a non-constant polynomial where $n \geq 1$ and $a_n \neq 0$. WLOG we may assume $a_n > 0$. Let $A = \max\{|a_0|, |a_1|, \dots, |a_{n-1}|\}$. For $x > 1$, by the triangle inequality, we have

$$\begin{aligned} |P(x)| &\geq a_n x^n - |a_{n-1}|x^{n-1} - |a_{n-2}|x^{n-2} - \cdots - |a_0| \\ &\geq a_n x^n - A x^{n-1} - A x^{n-2} - \cdots - A \\ &\geq a_n x^n - A x^{n-1} - A x^{n-1} - \cdots - A x^{n-1} \\ &= x^{n-1}(a_n x - A n). \end{aligned}$$

Since n, a_n and A are fixed, $|P(x)|$ can be arbitrarily large when x tends to infinity. This means $P(x)$ is unbounded.

12. **Solution 1:**

Let $P(x) = ax^2 + bx + c$. From the conditions, we need

$$\begin{aligned} a + b + c &= 2, \\ 4a + 2b + c &= 7, \\ 9a + 3b + c &= -1. \end{aligned}$$

The first equation gives $c = 2 - a - b$. Putting this into the other two equations, we get

$$\begin{aligned} 3a + b &= 5, \\ 8a + 2b &= -3. \end{aligned}$$

Therefore,

$$a = \frac{8a + 2b}{2} - (3a + b) = \frac{-3}{2} - 5 = -\frac{13}{2}.$$

Then we have $b = 5 - 3a = \frac{49}{2}$ and $c = 2 - a - b = -16$. Hence, the desired polynomial is

$$P(x) = -\frac{13}{2}x^2 + \frac{49}{2}x - 16.$$

Solution 2:

Using Lagrange interpolation formula, we have

$$\begin{aligned}
P(x) &= \frac{(x-2)(x-3)}{(1-2)(1-3)}P(1) + \frac{(x-1)(x-3)}{(2-1)(2-3)}P(2) + \frac{(x-1)(x-2)}{(3-1)(3-2)}P(3) \\
&= (x-2)(x-3) - 7(x-1)(x-3) - \frac{(x-1)(x-2)}{2} \\
&= -\frac{13}{2}x^2 + \frac{49}{2}x - 16.
\end{aligned}$$

13. Using Lagrange interpolation formula, we have

$$P(x) = \sum_{i=0}^n \left(\prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \right) P(x_i).$$

Since each $P(x_i) = y_i$ is rational and each x_i is rational, all coefficients are rational numbers.

14. WLOG we may assume the leading coefficient is positive. Thus, there exists a large integer N such that the equation $P(x) = N + k$ has a real solution for each integer $k = 0, 1, \dots, n$. From the condition, the solution x_k to $P(x) = N + k$ must be rational. Therefore, we can find rational numbers $x_0, x_1, \dots, x_n$ such that $P(x_k) = N + k$ for each k . Using Lagrange interpolation formula, $P(x)$ must have rational coefficients.
15. Let $Q(x) = (x+1)P(x) - x$. From the condition, we have $Q(k) = 0$ for $k = 0, 1, 2, \dots, n$. As $Q(x)$ is a polynomial of degree $n+1$, its $n+1$ roots are exactly $0, 1, 2, \dots, n$. Therefore,

$$Q(x) = cx(x-1)(x-2)\cdots(x-n)$$

for some constant c . Note that $Q(-1) = 0 - (-1) = 1$. Thus, we have

$$1 = c(-1)(-2)(-3)\cdots(-1-n).$$

This yields $c = \frac{(-1)^{n+1}}{(n+1)!}$. It follows that

$$P(x) = \frac{1}{x+1} \left(\frac{(-1)^{n+1}}{(n+1)!} x(x-1)(x-2)\cdots(x-n) + x \right)$$

and hence

$$P(n+1) = \frac{1}{n+2} \left(\frac{(-1)^{n+1}}{(n+1)!} (n+1)(n)(n-1)\cdots(1) + n+1 \right) = \frac{(-1)^{n+1} + n+1}{n+2}.$$

16. Consider

$$Q(x) = \sum_{r=0}^n \binom{x}{r}.$$

Note that $Q(x)$ is a polynomial of degree n . For $k = 0, 1, \dots, n$, we have

$$Q(k) = \sum_{r=0}^n \binom{k}{r} = \sum_{r=0}^k \binom{k}{r} = 2^k.$$

Thus, $Q(x)$ and $P(x)$ agree with each other in $n+1$ values. As they are polynomials of degree n , we have $P = Q$.

17. Using Lagrange interpolation formula, we have

$$P(x) = \sum_{k=0}^n \left(\prod_{\substack{j=0 \\ j \neq k}}^n \frac{x-j}{k-j} \right) P(k).$$

For each $k = 0, 1, \dots, n$, we have

$$\begin{aligned} \prod_{\substack{j=0 \\ j \neq k}}^n \frac{(n+1)-j}{k-j} &= \prod_{\substack{j=0 \\ j \neq k}}^n (n+1-j) \cdot \prod_{j=0}^{k-1} \frac{1}{k-j} \cdot \prod_{j=k+1}^n \frac{1}{k-j} \\ &= \frac{(n+1)!}{n+1-k} \times \frac{1}{k!} \times \frac{(-1)^{n-k}}{(n-k)!}. \end{aligned}$$

As

$$P(k) = \binom{n+1}{k}^{-1} = \frac{k!(n+1-k)!}{(n+1)!},$$

we find that

$$P(n+1) = \sum_{k=0}^n \frac{(n+1)!}{n+1-k} \times \frac{1}{k!} \times \frac{(-1)^{n-k}}{(n-k)!} \times \frac{k!(n+1-k)!}{(n+1)!} = \sum_{k=0}^n (-1)^{n-k}.$$

If n is odd, this yields $P(n+1) = 0$. If n is even, this yields $P(n+1) = 1$.

18. Using Lagrange interpolation formula, we have

$$P(x) = \sum_{k=0}^n \left(\prod_{\substack{j=0 \\ j \neq k}}^n \frac{x-j}{k-j} \right) P(k).$$

As $P(x)$ is monic, the leading coefficient is 1. This yields

$$1 = \sum_{k=0}^n \left(\prod_{\substack{j=0 \\ j \neq k}}^n \frac{1}{k-j} \right) P(k).$$

For each $k = 0, 1, \dots, n$, we have

$$\prod_{\substack{j=0 \\ j \neq k}}^n \frac{1}{k-j} = \prod_{j=0}^{k-1} \frac{1}{k-j} \cdot \prod_{j=k+1}^n \frac{1}{k-j} = \frac{1}{k!} \times \frac{(-1)^{n-k}}{(n-k)!}.$$

This implies

$$1 = \sum_{k=0}^n \frac{(-1)^{n-k}}{k!(n-k)!} P(k).$$

Suppose on the contrary that

$$|P(k)| < \frac{n!}{2^n}$$

for each $k = 0, 1, \dots, n$. Then, by the triangle inequality,

$$1 \leq \sum_{k=0}^n \frac{1}{k!(n-k)!} |P(k)| < \frac{1}{2^n} \sum_{k=0}^n \frac{n!}{k!(n-k)!} = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} = 1.$$

This is a contradiction. Thus, we have $|P(k)| \geq \frac{n!}{2^n}$ for some $k = 0, 1, \dots, n$.

19. Firstly, suppose $p(x)$ is divisible by m . Consider

$$\begin{aligned} p_1(x) &= p(x+1) - p(x) \\ &= (a_0(x+1)^k + a_1(x+1)^{k-1} + \dots + a_k) - (a_0x^k + a_1x^{k-1} + \dots + a_k) \\ &= a_0kx^{k-1} + \dots \end{aligned}$$

As $p(x+1)$ and $p(x)$ are divisible by m for any integer x , the polynomial $p_1(x)$ is divisible by m . Its leading coefficient is a_0k . Similarly, we can define $p_2(x) = p_1(x+1) - p_1(x)$, which is a polynomial divisible by m and its leading term is $a_0k(k-1)x^{k-2}$. Inductively, we define $p_j(x)$ for $j = 1, 2, \dots, k$. Finally, we obtain a polynomial $p_k(x) = a_0k!$ which is divisible by m . This implies $m \mid a_0k!$ as desired.

Conversely, suppose $m \mid a_0k!$. Define

$$p(x) = a_0x(x+1)\cdots(x+k-1).$$

For any integer x , $x(x+1)\cdots(x+k-1)$ is a product of k consecutive integers, and hence it is a multiple of $k!$. Thus $p(x)$ is a multiple of $a_0k!$ for any integer x . In particular, $m \mid p(x)$.

20. As x divides the right-hand side and $\gcd(x+2, x) = \gcd(2, x)$ is a constant, the linear factor x has to divide $P(x)$. Let $P(x) = xQ(x)$ for some polynomial $Q(x)$. Then the condition becomes $Q(x) = Q(x+2)$, and hence $Q(x) = Q(x+2) = Q(x+4) = \dots$. Note that a polynomial which has the same value at infinitely many points must be a constant polynomial. So $Q(x) = c$ and $P(x) = cx$ for some real number c .

Checking:

$$(x+2)P(x) = (x+2)(cx) = x(c(x+2)) = xP(x+2).$$

21. Let $Q(x) = P(x) - P(x-1)$. Then the equation becomes

$$Q(x+1) = Q(x) + 2.$$

Let $R(x) = Q(x+1) - Q(x)$. Then $R(x)$ is the constant polynomial 2.

If $P(x)$ is of degree n , then $Q(x)$ is of degree $n-1$ and $R(x)$ is of degree $n-2$. Therefore, we have $n = 2$.

Let $P(x) = ax^2 + bx + c$. Then

$$Q(x) = (ax^2 + bx + c) - (a(x-1)^2 + b(x-1) + c) = 2ax + b - a$$

and

$$R(x) = (2a(x+1) + b - a) - (2ax + b - a) = 2a.$$

As $R(x) = 2$, we have $a = 1$. Note that the steps are reversible. Therefore, $P(x)$ is any monic quadratic polynomial.

22. Clearly, $P(x)$ can be any constant polynomial. Suppose the degree of $P(x)$ is $n \geq 1$. Then the degree of $P(P(x))$ is n^2 and the degree of $P(x^2)$ is $2n$. Thus, we must have $n^2 = 2n$. The only possibility is $n = 2$.

Let $P(x) = ax^2 + bx + c$ where $a, b, c \in \mathbb{R}$ and $a \neq 0$. Then the equation becomes

$$a(ax^2 + bx + c)^2 + b(ax^2 + bx + c) + c = a(x^2)^2 + b(x^2) + c.$$

Comparing the coefficients, we need

$$\begin{aligned} a^2 &= a, \\ 2a^2b &= 0, \\ 2a^2c + ab^2 + ab &= b, \\ 2abc + b^2 &= 0, \\ ac^2 + bc + c &= c. \end{aligned}$$

From the first equation, as $a \neq 0$, we have $a = 1$. Then the second equation yields $b = 0$, while the third equation yields $c = 0$. The remaining two equations are satisfied in that case. Thus, $P(x) = x^2$ or it is a constant polynomial.

23. From $P(x) = Q(x)(P(x) + Q(x))$, we have $P(x)(1 - Q(x)) = Q(x)^2$. In particular, $1 - Q(x)$ is a factor of

$$Q(x)^2 = Q(x)^2 - 1 + 1 = (Q(x) - 1)(Q(x) + 1) + 1.$$

This means $1 - Q(x)$ divides 1, so that $Q(x)$ must be a constant polynomial.

Let $Q(x) = c$. Note that $c \neq 1$ from the equation. Thus,

$$P(x) = \frac{Q(x)^2}{1 - Q(x)} = \frac{c^2}{1 - c}.$$

So these are the solutions. One can check that $P(x) = Q(x)(P(x) + Q(x))$ holds for any constant $c \neq 1$.

24. If $n \geq 4$, then we have

$$x_1^2 + x_2^2 + \cdots + x_n^2 = a^2$$

and

$$x_1^4 + x_2^4 + \cdots + x_n^4 = a^4.$$

Squaring the first equation and taking the difference with the second, we obtain

$$2 \sum_{1 \leq i < j \leq n} x_i^2 x_j^2 = \left(\sum_{k=1}^n x_k^2 \right)^2 - \left(\sum_{k=1}^n x_k^4 \right) = 0.$$

As each term on the left is nonnegative, this implies at least $n-1$ variables x_j are equal to 0. WLOG assume $x_2 = x_3 = \cdots = x_n = 0$. Then the equation corresponding to $i = 1$ gives $x_1 = a$. Clearly, $(x_1, x_2, \dots, x_n) = (a, 0, 0, \dots, 0)$ and its permutations are solutions.

If $n = 3$, then we have

$$x_1 + x_2 + x_3 = a$$

and

$$x_1 x_2 + x_2 x_3 + x_3 x_1 = \frac{1}{2}((x_1 + x_2 + x_3)^2 - (x_1^2 + x_2^2 + x_3^2)) = 0.$$

Also, using the identity

$$x_1^3 + x_2^3 + x_3^3 - 3x_1 x_2 x_3 = (x_1 + x_2 + x_3)(x_1^2 + x_2^2 + x_3^2 - x_1 x_2 - x_2 x_3 - x_3 x_1),$$

we obtain

$$x_1 x_2 x_3 = \frac{1}{3}(a^3 - a \cdot a^2) = 0.$$

These show x_1, x_2, x_3 are the roots of $t^3 - at^2 = 0$. Thus, they are equal to $0, 0, a$ in some order.

If $n = 2$, then we have $x_1 + x_2 = a$ and $x_1 x_2 = \frac{1}{2}((x_1 + x_2)^2 - (x_1^2 + x_2^2)) = 0$. So $(x_1, x_2) = (0, a), (a, 0)$.

If $n = 1$, then clearly the solution is $x_1 = a$.

Therefore, the solutions are one of $x_1, x_2, \dots, x_n$ equals to a and the others equal to 0.

25. Prove or disprove the following.

(a) False. For example, for $P(x) = Q(x) = x$, we have $P(x) + Q(x) = 2x$. Then

$$\deg(P + Q) = 1 \neq 2 = 1 + 1 = \deg P + \deg Q.$$

(b) False. For example, for $P(x) = x$ and $Q(x) = -x + 1$, we have $P(x) + Q(x) = 1$. Then

$$\deg(P + Q) = 0 \neq 1 = \max\{\deg P, \deg Q\}.$$

(c) True. Let

$$\begin{aligned}P(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0, \\Q(x) &= b_m x^m + b_{m-1} x^{m-1} + \cdots + b_0\end{aligned}$$

where $a_n, b_m \neq 0$. Then

$$PQ(x) = a_n b_m x^{n+m} + (a_n b_{m-1} + a_{n-1} b_m) x^{n+m-1} + \cdots + a_0 b_0.$$

This shows

$$\deg(PQ) = n + m = \deg P + \deg Q.$$

(d) False. For example, for $P(x) = x^3$, we have $P(x^2) = (x^2)^3 = x^6$. Then

$$\deg P(x^2) = 6 \neq 9 = 3^2 = (\deg P)^2.$$

The correct relation is $\deg P(x^2) = 2 \deg P$.

(e) True. Note that $f(x) = x$ for all positive reals x . This shows the function $f(x) - x$ has infinitely many roots. However, it is not identically zero. This is a contradiction as a nonzero polynomial only has finitely many roots.

(f) False. For example, all the polynomials $P(x) = ax$ satisfy $P(0) = 0$ for any nonzero $a \in \mathbb{R}$. In general, we need $n + 1$ points to determine the unique polynomial of degree at most n (not necessarily equal to n).

Identity

26. (a) We have $a^2 - b^2 = (a - b)(a + b)$.

(b) We have $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$.

(c) We have $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \cdots + b^{n-1})$.

(d) We have $a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \cdots + b^{n-1})$.

(e) We have $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.

(f) We have

$$\begin{aligned}a^4 + 4b^4 &= a^4 + 4a^2b^2 + 4b^4 - 4a^2b^2 = (a^2 + 2b^2)^2 - (2ab)^2 \\&= (a^2 + 2b^2 - 2ab)(a^2 + 2b^2 + 2ab).\end{aligned}$$

(g) We have $(ab + 1)^2 + (a - b)^2 = a^2b^2 + a^2 + b^2 + 1 = (a^2 + 1)(b^2 + 1)$.

- (h) If we put $a = b$, then the expression equals 0. Thus, $a - b$ is a factor. Similarly, each of $b - c$ and $c - a$ is a linear factor. This gives

$$a^3b + b^3c + c^3a - ab^3 - bc^3 - ca^3 = (a - b)(b - c)(c - a)f(a, b, c).$$

Note that the remaining factor $f(a, b, c)$ is linear. Let $f(a, b, c) = \alpha a + \beta b + \gamma c$. By checking the coefficients of the term a^3b , we find that $\alpha = -1$. Similarly, we have $\beta = \gamma = -1$. Therefore,

$$a^3b + b^3c + c^3a - ab^3 - bc^3 - ca^3 = -(a - b)(b - c)(c - a)(a + b + c).$$

- (i) We have

$$\begin{aligned} (a + 1)(a + 2)(a + 3)(a + 4) + 1 &= ((a + 1)(a + 4))((a + 2)(a + 3)) + 1 \\ &= (a^2 + 5a + 4)(a^2 + 5a + 6) + 1 \\ &= (a^2 + 5a + 5 - 1)(a^2 + 5a + 5 + 1) + 1 \\ &= (a^2 + 5a + 5)^2 - 1 + 1 \\ &= (a^2 + 5a + 5)^2. \end{aligned}$$

27. (a) We have

$$a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = S_1^2 - 2S_2.$$

- (b) We have

$$\begin{aligned} a^3 + b^3 + c^3 &= a^3 + b^3 + c^3 - 3abc + 3abc \\ &= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) + 3abc \\ &= S_1(S_1^2 - 3S_2) + 3S_3. \end{aligned}$$

- (c) We have

$$\begin{aligned} a^2b + a^2c + b^2c + b^2a + c^2a + c^2b &= (a + b + c)(ab + bc + ca) - 3abc \\ &= S_1S_2 - 3S_3. \end{aligned}$$

- (d) We have

$$a^2bc + b^2ca + c^2ab = (a + b + c)abc = S_1S_3.$$

- (e) We have

$$a^2b^2 + b^2c^2 + c^2a^2 = (ab + bc + ca)^2 - 2(a^2bc + b^2ca + c^2ab) = S_2^2 - 2S_1S_3.$$

- (f) We have

$$\begin{aligned} &a^3b + a^3c + b^3c + b^3a + c^3a + c^3b \\ &= (a^2 + b^2 + c^2)(ab + bc + ca) - (a^2bc + b^2ca + c^2ab) \\ &= (S_1^2 - 2S_2)S_2 - S_1S_3 \\ &= S_1^2S_2 - 2S_2^2 - S_1S_3. \end{aligned}$$

(g) We have

$$\begin{aligned} a^4 + b^4 + c^4 &= (a^2 + b^2 + c^2)^2 - 2(a^2b^2 + b^2c^2 + c^2a^2) \\ &= (S_1^2 - 2S_2)^2 - 2(S_2^2 - 2S_1S_3) \\ &= S_1^4 - 4S_1^2S_2 + 2S_2^2 + 4S_1S_3. \end{aligned}$$

28. Putting all terms to the left-hand side, the equation is equivalent to

$$\frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2) = 0.$$

Therefore, we must have $a - b = b - c = c - a = 0$. This implies $a = b = c$. Clearly, this is a solution.

29. Let $S = a^3b + b^3c + c^3a$. Note that

$$a^3b + b^3c + c^3a - ab^3 - bc^3 - ca^3 = -(a-b)(b-c)(c-a)(a+b+c) = 0.$$

Therefore, $ab^3 + bc^3 + ca^3 = S$. Consider

$$2S = a^3b + b^3c + c^3a + ab^3 + bc^3 + ca^3.$$

We have

$$2S = (a^2 + b^2 + c^2)(ab + bc + ca) - abc(a + b + c) = 6(ab + bc + ca)$$

where

$$ab + bc + ca = \frac{1}{2}((a+b+c)^2 - (a^2 + b^2 + c^2)) = -3.$$

$$\text{Thus, } S = \frac{6(-3)}{2} = -9.$$

30. The given equation is equivalent to

$$(bc + ca + ab)(a + b + c) = abc.$$

After expansion, this becomes

$$a^2b + a^2c + b^2c + b^2a + c^2a + c^2b + 2abc = 0.$$

Thus,

$$(a+b)(b+c)(c+a) = a^2b + a^2c + b^2c + b^2a + c^2a + c^2b + 2abc = 0.$$

31. The given equation can be transformed to

$$(a+b)(b+c)(c+a) = 0.$$

WLOG we may assume $a + b = 0$. Then

$$\frac{1}{a^n} + \frac{1}{b^n} + \frac{1}{c^n} = \frac{1}{a^n} - \frac{1}{a^n} + \frac{1}{c^n} = \frac{1}{c^n} = \frac{1}{a^n - a^n + c^n} = \frac{1}{a^n + b^n + c^n}$$

since n is odd.

32. Define $x_{n+1} = x_1$ and $x_{n+2} = x_2$. For each $k = 1, 2, \dots, n$, we have

$$\frac{x_k x_{k+2} - x_{k+1}^2}{(x_k + x_{k+1})(x_{k+1} + x_{k+2})} = \frac{x_k}{x_k + x_{k+1}} - \frac{x_{k+1}}{x_{k+1} + x_{k+2}}.$$

Summing over $k = 1, 2, \dots, n$, we obtain

$$\begin{aligned} \sum_{k=1}^n \frac{x_k x_{k+2} - x_{k+1}^2}{(x_k + x_{k+1})(x_{k+1} + x_{k+2})} &= \sum_{k=1}^n \frac{x_k}{x_k + x_{k+1}} - \sum_{k=1}^n \frac{x_{k+1}}{x_{k+1} + x_{k+2}} \\ &= \sum_{k=1}^n \frac{x_k}{x_k + x_{k+1}} - \sum_{k=1}^n \frac{x_k}{x_k + x_{k+1}} = 0 \end{aligned}$$

by shifting the indices of the second summation.

33. We shall prove by induction that

$$\begin{aligned} &1 + \frac{1}{x_1} + \frac{x_1 + 1}{x_1 x_2} + \frac{(x_1 + 1)(x_2 + 1)}{x_1 x_2 x_3} + \dots + \frac{(x_1 + 1)(x_2 + 1) \cdots (x_{n-1} + 1)}{x_1 x_2 \cdots x_n} \\ &= \frac{(x_1 + 1)(x_2 + 1) \cdots (x_n + 1)}{x_1 x_2 \cdots x_n} \end{aligned} \quad (1)$$

for any positive integer n . When $n = 1$, we have $1 + \frac{1}{x_1} = \frac{x_1 + 1}{x_1}$. Assume

$$\begin{aligned} &1 + \frac{1}{x_1} + \frac{x_1 + 1}{x_1 x_2} + \frac{(x_1 + 1)(x_2 + 1)}{x_1 x_2 x_3} + \dots + \frac{(x_1 + 1)(x_2 + 1) \cdots (x_{k-1} + 1)}{x_1 x_2 \cdots x_k} \\ &= \frac{(x_1 + 1)(x_2 + 1) \cdots (x_k + 1)}{x_1 x_2 \cdots x_k} \end{aligned}$$

for some $k \geq 1$. Then we have

$$\begin{aligned} &1 + \frac{1}{x_1} + \frac{x_1 + 1}{x_1 x_2} + \frac{(x_1 + 1)(x_2 + 1)}{x_1 x_2 x_3} + \dots + \frac{(x_1 + 1)(x_2 + 1) \cdots (x_k + 1)}{x_1 x_2 \cdots x_{k+1}} \\ &= \frac{(x_1 + 1)(x_2 + 1) \cdots (x_k + 1)}{x_1 x_2 \cdots x_k} + \frac{(x_1 + 1)(x_2 + 1) \cdots (x_k + 1)}{x_1 x_2 \cdots x_{k+1}} \\ &= \frac{(x_1 + 1)(x_2 + 1) \cdots (x_k + 1)(x_{k+1} + 1)}{x_1 x_2 \cdots x_{k+1}}. \end{aligned}$$

This establishes the case $n = k + 1$. By induction, the identity (1) holds for all $n \geq 1$. Thus, the equation can be rewritten as

$$\frac{(x_1 + 1)(x_2 + 1) \cdots (x_n + 1)}{x_1 x_2 \cdots x_n} = 0.$$

The solutions are obviously $x_j = -1$ for at least one j (and $x_1, x_2, \dots, x_n \neq 0$).

34. We shall prove by induction that

$$\begin{aligned} & \frac{1}{1+a_1} + \frac{a_1}{(1+a_1)(1+a_2)} + \cdots + \frac{a_1 a_2 \cdots a_{n-1}}{(1+a_1)(1+a_2) \cdots (1+a_n)} \\ &= 1 - \frac{a_1 a_2 \cdots a_n}{(1+a_1)(1+a_2) \cdots (1+a_n)} \end{aligned} \quad (1)$$

for any positive integer n . When $n = 1$, we have $\frac{1}{1+a_1} = 1 - \frac{a_1}{1+a_1}$. Assume

$$\begin{aligned} & \frac{1}{1+a_1} + \frac{a_1}{(1+a_1)(1+a_2)} + \cdots + \frac{a_1 a_2 \cdots a_{k-1}}{(1+a_1)(1+a_2) \cdots (1+a_k)} \\ &= 1 - \frac{a_1 a_2 \cdots a_k}{(1+a_1)(1+a_2) \cdots (1+a_k)} \end{aligned}$$

for some $k \geq 1$. Then we have

$$\begin{aligned} & \frac{1}{1+a_1} + \frac{a_1}{(1+a_1)(1+a_2)} + \cdots + \frac{a_1 a_2 \cdots a_k}{(1+a_1)(1+a_2) \cdots (1+a_{k+1})} \\ &= 1 - \frac{a_1 a_2 \cdots a_k}{(1+a_1)(1+a_2) \cdots (1+a_k)} + \frac{a_1 a_2 \cdots a_k}{(1+a_1)(1+a_2) \cdots (1+a_{k+1})} \\ &= 1 - \frac{a_1 a_2 \cdots a_k (1+a_{k+1} - 1)}{(1+a_1)(1+a_2) \cdots (1+a_{k+1})} \\ &= 1 - \frac{a_1 a_2 \cdots a_{k+1}}{(1+a_1)(1+a_2) \cdots (1+a_{k+1})}. \end{aligned}$$

This establishes the case $n = k + 1$. By induction, the identity (1) holds for all $n \geq 1$. As the variables are nonnegative, it is now obvious that the expression is at most 1, with equality case $a_j = 0$ for some j .

Root and Vieta's Theorem

35. By Vieta's theorem, we have

$$\begin{aligned} \alpha + \beta + \gamma &= \frac{9}{2}, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= \frac{1}{2}, \\ \alpha\beta\gamma &= -\frac{11}{2}. \end{aligned}$$

(a) We have

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = \left(\frac{9}{2}\right)^2 - 2\left(\frac{1}{2}\right) = \frac{77}{4}.$$

(b) We have

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = -\frac{1}{11}.$$

(c) Since α is a root of the equation, we have

$$2\alpha^3 - 9\alpha^2 + \alpha + 11 = 0.$$

Therefore,

$$\begin{aligned} 2\alpha^3 - 6\alpha^2 + \alpha + 3\beta^2 + 3\gamma^2 &= (9\alpha^2 - \alpha - 11) - 6\alpha^2 + \alpha + 3\beta^2 + 3\gamma^2 \\ &= 3(\alpha^2 + \beta^2 + \gamma^2) - 11 = \frac{187}{4}. \end{aligned}$$

36. By Vieta's theorem, we have

$$\begin{aligned} \alpha + \beta + \gamma &= 3, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= -1, \\ \alpha\beta\gamma &= -2. \end{aligned}$$

From these, we find that

$$(\alpha\beta)(\beta\gamma) + (\beta\gamma)(\gamma\alpha) + (\gamma\alpha)(\alpha\beta) = (\alpha + \beta + \gamma)\alpha\beta\gamma = -6$$

and

$$(\alpha\beta)(\beta\gamma)(\gamma\alpha) = (\alpha\beta\gamma)^2 = 4.$$

Together with $\alpha\beta + \beta\gamma + \gamma\alpha = -1$, the monic cubic polynomial with roots $\alpha\beta, \beta\gamma, \gamma\alpha$ is

$$x^3 + x^2 - 6x - 4$$

by Vieta's theorem.

37. Suppose on the contrary that all the roots are rational. Note that $a \neq 0$. By the rational root theorem, the denominator of each root divides the leading coefficient a . Let the roots be $\frac{\alpha}{a}, \frac{\beta}{a}, \frac{\gamma}{a}$ where α, β, γ are integers. By Vieta's theorem, we have

$$\begin{aligned} \frac{\alpha}{a} + \frac{\beta}{a} + \frac{\gamma}{a} &= -\frac{b}{a}, \\ \frac{\alpha}{a} \cdot \frac{\beta}{a} + \frac{\beta}{a} \cdot \frac{\gamma}{a} + \frac{\gamma}{a} \cdot \frac{\alpha}{a} &= \frac{c}{a}, \\ \frac{\alpha}{a} \cdot \frac{\beta}{a} \cdot \frac{\gamma}{a} &= -\frac{d}{a}. \end{aligned}$$

These can be rewritten as

$$\begin{aligned} \alpha + \beta + \gamma &= -b, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= ca, \\ \alpha\beta\gamma &= -da^2. \end{aligned}$$

As a and d are odd, the last equation shows that α, β, γ are odd. Then the first equation shows that b is odd, while the second equation shows that c is odd. This contradicts the fact that bc is even. Therefore, one of the roots is irrational.

38. By Vieta's theorem, we have

$$\begin{aligned}a + b + c &= -a, \\ab + bc + ca &= b, \\abc &= -c.\end{aligned}$$

From the third equation, we have $c = 0$ or $ab = -1$.

If $c = 0$, then the second equation becomes $ab = b$. Hence, $b = 0$ or $a = 1$. For $b = 0$, the first equation gives $a = 0$. Clearly, $a = b = c = 0$ is a solution. For $a = 1$, the first equation gives $b = -2a = -2$. The triple $(a, b, c) = (1, -2, 0)$ is a solution since all three equations are satisfied.

If $ab = -1$, as a, b are integers, we have $(a, b) = (1, -1)$ or $(-1, 1)$. For the former case, the first equation gives $c = -2a - b = -1$. One checks that

$$ab + bc + ca = -1 = b.$$

For the latter case, the first equation gives $c = -2a - b = 1$. However, in that case we get the contradiction

$$ab + bc + ca = -1 \neq b.$$

Therefore, the solutions are $(a, b, c) = (0, 0, 0), (1, -2, 0), (1, -1, -1)$.

39. We have

$$xy + yz + zx = \frac{1}{2}((x + y + z)^2 - (x^2 + y^2 + z^2)) = \frac{1}{2}(3^2 - 3) = 3$$

and

$$\begin{aligned}xyz &= \frac{1}{3}((x^3 + y^3 + z^3) - (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)) \\&= \frac{1}{3}(3 - 3(3 - 3)) = 1.\end{aligned}$$

Together with $x + y + z = 3$, by Vieta's theorem, x, y, z are the roots of

$$t^3 - 3t^2 + 3t - 1.$$

As this polynomial can be factorized as $(t - 1)^3$, the three roots are $x = y = z = 1$, which obviously satisfy the given equations.

40. Let $xy + yz + zx = a$. Then

$$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx) = 9 - 2a.$$

Let $xyz = b$. Using the identity

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx),$$

we have $15 - 3b = 3(9 - 3a)$. This gives $b = 3a - 4$.

Next, we find that

$$x^2y^2 + y^2z^2 + z^2x^2 = (xy + yz + zx)^2 - 2xyz(x + y + z) = a^2 - 6b = a^2 - 18a + 24.$$

Using the identity

$$(x^2 + y^2 + z^2)^2 = (x^4 + y^4 + z^4) + 2(x^2y^2 + y^2z^2 + z^2x^2),$$

we have $(9 - 2a)^2 = 35 + 2(a^2 - 18a + 24)$, i.e. $2a^2 = 2$. This gives $a = \pm 1$. As

$$9 - 2a = x^2 + y^2 + z^2 < 10,$$

the only solution is $a = 1$. In that case, we have

$$\begin{aligned} x + y + z &= 3, \\ xy + yz + zx &= 1, \\ xyz &= -1. \end{aligned}$$

By Vieta's theorem, x, y, z are the roots of $t^3 - 3t^2 + t + 1 = 0$. One checks that all the conditions are satisfied in that case.

Now, we have $x^5 = x^2(x^3) = x^2(3x^2 - x - 1) = 3x^4 - x^3 - x^2$. Thus,

$$x^5 + y^5 + z^5 = 3(x^4 + y^4 + z^4) - (x^3 + y^3 + z^3) - (x^2 + y^2 + z^2) = 83.$$

41. Let $p(x) = x^3 + ax^2 + bx + c$ where $a, b, c \in \mathbb{Z}$. Let the roots be t, u, tu . By Vieta's theorem, we have

$$\begin{aligned} t + u + tu &= -a, \\ tu + t^2u + tu^2 &= b, \\ t^2u^2 &= -c. \end{aligned}$$

From the last equation, we must have $-c \geq 0$. Then we have $tu = \pm\sqrt{-c}$. The second equation becomes

$$b = tu(1 + t + u) = \pm\sqrt{-c}(1 + t + u).$$

Thus, either $c = 0$ (in which case $b = 0$) or $t + u = \pm \frac{b}{\sqrt{-c}} - 1$. In the latter case, the first equation becomes

$$-a = t + u + tu = \pm \frac{b}{\sqrt{-c}} - 1 \pm \sqrt{-c}.$$

This means $b - c = \pm \sqrt{-c}(1 - a)$. Therefore, either $b = c$ (which includes the case $b = c = 0$) or $\sqrt{-c}$ is a rational number. The latter case only holds when $\sqrt{-c}$ is an integer since c is an integer. Thus, the integer $b - c$ is divisible by the integer $1 - a$ in any case (including the case $b = c$).

Now, we have $2p(-1) = 2(a - b + c - 1)$ and

$$p(1) + p(-1) - 2(1 + p(0)) = (a + b + c + 1) + (a - b + c - 1) - 2(1 + c) = 2(a - 1).$$

Therefore, $p(1) + p(-1) - 2(1 + p(0)) \mid 2p(-1)$ if and only if $a - 1 \mid a - b + c - 1$, which is the same as $a - 1 \mid c - b$. This holds from above.

42. Suppose on the contrary that there exists a root $t \leq 0$. Then we have

$$t^3 \leq 0, -at^2 \leq 0, bt \leq 0, -c < 0$$

since $a, b, c > 0$. Thus, we get

$$t^3 - at^2 + bt - c < 0.$$

This is a contradiction. Thus, all the roots are positive.

43. Let $x_n = x_0 + na$ and $y_n = y_0 + nb$ for some real constants x_0, a, y_0, b . We have $P(x_0 + na) = y_0 + nb$ for infinitely many n 's. Using the change of variable $x = x_0 + na$, we get

$$P(x) = y_0 + \left(\frac{x - x_0}{a}\right)b = \frac{b}{a}x + y_0 - \frac{b}{a}x_0.$$

Since $a \neq 0$, this holds for infinitely many x 's. As $P(x)$ is a polynomial, this holds for all real numbers x . As $b \neq 0$, $P(x)$ has degree 1, i.e. it is a linear polynomial.

44. Consider the equation $P(x) = \sqrt{x}$. Squaring both sides, we obtain $P(x)^2 = x$, i.e. $P(x)^2 - x = 0$.

We may assume $P(x)$ is a non-constant polynomial. Let $n \geq 1$ be the degree of $P(x)$. Then $P(x)^2 - x$ is a polynomial of degree $2n$. Therefore, $P(x)^2 - x = 0$ has at most $2n$ solutions. In other words, only finitely many x may satisfy $P(x) = \sqrt{x}$.

45. Let $P(x) = x^3 + 4x^2 + x - 1$. Note that

$$P(-1) = 1, P(0) = -1, P(1) = 5.$$

By the intermediate value theorem, there exists a root a in the range $-1 < a < 0$ and another root b in the range $0 < b < 1$. Also, since $P(x)$ is negative when x tends to $-\infty$, there is a third root c in the range $c < -1$. Thus, there are three distinct real roots.

46. Let $P(x) = x(x-2)(x-4)(x-6) + (x-1)(x-3)(x-5)(x-7)$. Note that

$$P(1) = (1)(-1)(-3)(-5) < 0,$$

$$P(3) = (3)(1)(-1)(-3) > 0,$$

$$P(5) = (5)(3)(1)(-1) < 0.$$

By the intermediate value theorem, there exists a root a in the range $1 < a < 3$ and another root b in the range $3 < b < 5$. Also, note that the leading coefficient of $P(x)$ is positive and the degree is 4. Therefore, $P(x)$ is positive when x tends to ∞ or $-\infty$. Hence, there is a third root c in the range $c < 1$ and a fourth root d in the range $d > 5$. Thus, there are four distinct real roots.

47. WLOG, assume $a_1 < a_2 < \dots < a_k < 0 \leq a_{k+1} < a_{k+2} < \dots < a_n$ for some integer $k = 0, 1, \dots, n$. Let

$$f(x) = \frac{a_1}{a_1 - x} + \frac{a_2}{a_2 - x} + \dots + \frac{a_n}{a_n - x} - n.$$

When x is slightly less than a_1 , $f(x)$ is dominated by the term $\frac{a_1}{a_1 - x}$, which is negative. This shows

$$\lim_{x \rightarrow a_1^-} f(x) = -\infty.$$

When x is slightly larger than a_1 , $f(x)$ is dominated by the same term, but the sign changes. Indeed, we have

$$\lim_{x \rightarrow a_1^+} f(x) = \infty.$$

Similary, we have

$$\lim_{x \rightarrow a_2^-} f(x) = -\infty.$$

As $f(x) > 0$ for x slightly larger than a_1 and $f(x) < 0$ for x slightly smaller than a_2 , there must be a root in the interval (a_1, a_2) by the intermediate value theorem as $f(x)$ is continuous on the interval (a_1, a_2) . Similarly, there is a root in each of the intervals $(a_2, a_3), \dots, (a_{k-1}, a_k)$.

Using similar arguments in the positive intervals, we can show that there is a root in each of the intervals $(a_{k+1}, a_{k+2}), \dots, (a_{n-1}, a_n)$.

Also, it is obvious that $f(0) = 1 + 1 + \dots + 1 - n = 0$. This shows there is another root $x = 0$ in the interval $(a_k, a_{k+1}]$.

We have shown that there is a root in the interval bounded by two consecutive numbers among the a_j 's (the same also holds when $k = 0$ and $k = n$). Thus, there are at least $n - 1$ roots to the equation $f(x) = 0$.

48. Suppose on the contrary that $P(x)$ has a real root a_0 . As $P(x)$ divides $P(x^2 + x + 1)$, we have $P(a_0^2 + a_0 + 1) = 0$. This shows $a_1 = a_0^2 + a_0 + 1$ is a real root of $P(x)$. Note that

$$a_1 = a_0^2 + a_0 + 1 > a_0.$$

By repeating the same process, we can find real numbers $a_2, a_3, \dots$ such that each a_j is a root of $P(x)$ and such that $a_0 < a_1 < a_2 < \dots$. This is a contradiction since a nonzero polynomial cannot have infinitely many roots. Thus, $P(x)$ has no real roots.